

A2 FILE CMD

User guide · 0.9.4

DOS3.3: the essentials · ProDOS: the complete workspace

From 48 KB and 40 columns to 128 KB and 80 columns



A2 File Cmd ProDOS XL 0.9.4 in 80 columns: two panels captured in POM2

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A2 File Cmd DOS3.3

Two panels, a source and a destination: DOS3.3 distils A2 File Cmd into selecting, reading, editing and verified copying. It uses an **Apple II+ with 48 KB**, NMOS 6502, 40 columns and **two Disk II drives on the boot controller**. The following chapters extend this workflow with ProDOS folders and tools.

Boot A2FILECMD-DOS3.3-0.9.4.dsk, or use **BRUN A2FC** from DOS 3.3. Both panels start on the boot disk; each remembers drive, selection and scroll. Catalogs hold up to 105 files and show 19 rows. ? opens help.

Keys	Action
TAB/Ctrl-I or 1; / or 4	Switch panels; change active drive.
Up/Down or I/K; Left/Right	Previous/next file; previous/next page. [/] first/last.
Return or 2; T/H/G; B	Open; force text/hex/hi-res; confirm BRUN.
Space; Ctrl-T/N; *	Tag one; all/none; invert tags.
C or 3; N/E; D/R/L	Copy; new/edit text; delete/rename/lock.
Ctrl-R or 5; =	Reread both panels; same disk opposite.
? or 6; Q or 7; Escape	Help; confirm quit; leave preview/help/editor.

Questions use **Y** to confirm and **N/Escape** to cancel. Other keys do not confirm. Escape in the browser has no effect: DOS has no parent directories. After changing a disk, reread with Ctrl-R. Prompts and the result occupy the footer; read them before inserting another disk.

Copy and edit

1. Put the panels on different drives using TAB and /.
2. Select/tag the sources and press C. Check COPY name? or the marked count.
3. Confirm Y, then wait for the progress bar and final result. **Once started, the disk-writing operation cannot be interrupted from the keyboard.**

Copies preserve DOS names, type, lock flag and all data-sector bytes. Existing names are skipped. Destination sectors are reserved first, written and read back, then the catalog entry is published. An uncertain write stops the batch; the source is neither written nor deleted. There is no single-drive copy by swapping disks.

N/E uses the same exclusive writer. E accepts at most **8 KB** and refuses larger files rather than truncating them. It always saves to a new name. R refuses an existing name or locked file. D skips locked files; L toggles the cursor's lock, or unlocks a tagged set if any member is locked, otherwise locks the set.

Preview and limits

Return identifies text, hi-res or a binary program from content and DOS headers; a program asks BRUN NAME?. T/H previews only the first 256 stored bytes; G shows the first 8 KB as hi-res. The hex preview shows eight bytes a row with their characters, one half of the sector at a time: Left/Right (or -/+ , </>) switch between bytes 00-7F and 80-FF. A valid 8 KB BIN load at \$2000/\$4000 is a picture; unsupported or unsafe program headers fall back to hex. BRUN leaves A2FC.

Only standard 35-track, 16-sector DOS 3.3 is supported. Sparse/inconsistent chains, reserved-track data, 13-sector, 40-track and protected formats are refused. Before a write, the catalog entry must still match the panel's identity; a changed disk is refused, though a byte-identical twin cannot be distinguished. Physical write protection is checked. After an uncertain error, reserved space is retained and further writes are blocked for that run: preserve and inspect a copy of the disk. Readback cannot repair metadata damaged by a power failure.

From DOS3.3 to ProDOS

ProDOS builds on the same two-panel workflow: folders, attributes, archives, documents, pictures, music and disk tools. It needs an **Apple IIe, //c or IIgs with 128 KB and 80 columns**. This is a separate program: its commands, plugins and recovery filenames do not apply to the DOS3.3 edition introduced first.

Five self-contained images

Image	Choose it for
A2FILECMD-DOS3.3-0.9.4.dsk	Standalone DOS 3.3 file manager for Apple II+ and two Disk II drives.
A2FILECMD-PRODOS-140K-0.9.4.dsk	One 5¼-inch ProDOS disk with essential file operations and text editing; any IIe.
A2FILECMD-PRODOS-800K-0.9.4.po	All ProDOS tools and BASIC runtimes, without the demonstration corpus; any IIe.
A2FILECMD-PRODOS-XL-0.9.4.2mg	Complete 32 MB ProDOS edition with tools and demonstrations; any IIe.
A2FILECMD-PRODOS-XL-65C02-enhanced-0.9.4.2mg	Complete XL for an enhanced IIe, //c or IIgs: MouseText and optional mouse support.

The choice is **enhanced or not**. The enhanced XL needs an enhanced IIe, a //c or a IIgs; every other ProDOS image runs on any IIe, enhanced or not. An enhanced machine always has the 65C02 the enhanced XL is compiled for: the launcher checks machine, processor and memory before starting, so a hand-modified machine is refused rather than crashed. Mouse hardware is optional.

140K and DOS3.3 use DOS-sector-order .dsk; 800K uses ProDOS-order .po. XL uses .2mg. Renaming an extension does not convert an image. With all release images and the PDF in the same directory, check their downloads:

```
sha256sum -c SHA256SUMS-0.9.4.txt
```

Install and start

Boot the selected image, or launch A2FILE.SYSTEM from a ProDOS selector. When installing elsewhere, keep A2FILE.SYSTEM beside the complete A2FILE/ directory and copy RECOVER too. Keep the program and its native plugins from the same build and CPU edition; do not mix them.

The 140K disk provides navigation, tags, copy/move/delete, rename and attributes, text editing, text/hex readers, comparison, formatting, verification and ProDOS image extraction. Advanced disk tools, DOS extraction, archives, media players and BASIC runtimes require **800K or XL**. Their menus list the available tools without asking for category disks. Both complete editions include all 82 overlays; only XL includes DEMO/: one example of every kind of file A2 File Cmd opens, in DOCUMENTS, PICTURES (with an HGR album), MUSIC, ARCHIVES, DISKS, PROGRAMS and FONTS.SHAPES, and a README saying what is where.

Before working

Use copies of valuable disks. Read each target name before confirming an operation. Keep space for both old and new versions during replacement. Do not remove a disk until the operation reports its result.

Some tools use auxiliary memory and rebuild **/RAM empty**. A2FC asks before that use and explicitly warns that **ALL /RAM files will be lost**. Copy them elsewhere first; declining preserves them. This applies to affected picture viewers, ShrinkIt, disk-image operations, Disk II formatting, NIBCOPY and full large-volume checks. Ordinary text and music readers preserve /RAM.

Keys and panels

TAB changes the active panel. The highlighted row is the cursor; * marks a tagged entry and L a locked one. Directories end in /. The header's star shows the sort order. Messages and progress appear below the panels.

Keys	Action
Up/Down; Left/Right or < / >	Move one entry; move one page. [/] goes to first/last.
Return / Escape //	Open; parent directory; list volumes. Return asks before running a program.
TAB / = / Ctrl-R	Other panel; same directory opposite; reread both panels.
Space / * / Ctrl-T / Ctrl-N	Tag one; invert tags; tag all; clear tags.
S / M / ' then initial	Change sort; tag size/date differences; jump to a name.
C / V / D	Copy; move; delete. Use tags, or the cursor when none are tagged.
R / K / A / L	Rename; new directory; type/auxtype; lock/unlock.
T / H / I / E / X	Text; hex; picture; edit; run after confirmation.
W / F / ! / ? / Q	Disk images; format; tools menu; help; quit after confirmation.
1 ... 0	Activate the corresponding bottom key-bar button.

Names start with a letter, then letters, digits or periods, up to 15 characters. Directories can be tagged; . . cannot. Returning to a parent selects the child you left if it still exists. Large directories use unsorted windows in disk order; continue through the window edge to reach later entries.

Enhanced mouse support lets you select a row, click it again to open, click a path to go up, a header to sort or a key-bar button to act. Prompts, viewers and the editor use the keyboard. On quit, A2FC saves panel paths, sorting and active side; a failed preference save lets you cancel quitting.

Copy, move and edit

1. Open the destination in the other panel. In the source, clear unwanted tags with **Ctrl-N**, then select or tag the entries to transfer.
2. Press **C** or **V**. For an existing target: **O** overwrites, **S** skips, **A** overwrites all and **N** overwrites none. Check the destination first.
3. Wait for **Copying** and **Verifying**, then read the result. Move removes a source only after its copy is verified. Existing directories are filled in; copying into the source or its descendants is refused.

Copy preserves type and auxtype. Replacements retain the old target during verification. **Escape** interrupts; completed work remains. If cleanup fails, a temporary can remain: use the recovery procedure before retrying. Deep trees may be refused; process smaller subdirectories rather than bypassing the limit.

! → Files → MOVE can relocate a single entry within a volume without copying its data blocks. A marked batch first reserves A2MOVE.LST, refuses marked directories and stops on error or Escape. Cross-volume copies are verified before source deletion. Keep any retained work list for review.

E edits up to **5,104 bytes**, with CR line endings; loading strips the high bit and long lines do not wrap. On a directory or . . , E creates text. Arrows move; Delete/Ctrl-D erase left/right; Ctrl-A/E goes to line start/end; Ctrl-P/N changes page; Ctrl-T/B goes to text start/end. Return splits a line; Tab inserts four spaces. Escape opens **S** save, **X** save/exit, **Q** abandon (confirm changes), or Escape to continue. Saving preserves type/auxtype and uses verified A2FC.EDIT plus A2FC.ED.BAK; existing recovery files block saving.

Recover after an incident

Every ProDOS disk includes **RECOVER** at its root. Open it with Return; the ? help screen points to it. Recovery begins by preserving evidence, not by retrying the failed operation or deleting the files it left behind.

Preserve, identify, recover

1. Stop and record the exact error, source path and intended destination. Make a disk image or duplicate before changing anything. Work on the duplicate and recover onto a separate healthy disk. If reads are unreliable and no independent copy is possible, stop and seek help.
2. Keep the target, temporary and backup. A name, date or matching size does not prove completeness. A generic A2FC.BAK may belong to an earlier operation: if its original target is unknown, preserve it without guessing.
3. On the **duplicate only**, rename one identified candidate to an unused RECOV.OLD or RECOV.NEW. This is necessary for A2FC.COPY, which C cannot copy under that reserved name. Choose another name if it is already taken.
4. Put a fresh recovery directory in the other panel, clear source tags with **Ctrl-N**, select the candidate and press **C**. Refuse overwrite. Wait for copying **and verification**; an error means recovery is not confirmed.
5. Check the recovered content with its reader, or compare all bytes against a known good source. Do not execute an unknown program to test it. Only after validation, give the recovered copy its intended name in an empty directory. Keep the disk image and other candidates until satisfied.

Recognize the files

Operation	New candidate	Previous version
Copy	A2FC.COPY	A2FC.BAK
Editor	A2FC.EDIT	A2FC.ED.BAK
SYNC	A2FC.SYNC	A2FC.BAK
Text / image conversion	TXTCONV.TMP / IMGCONV.TMP	A2FC.BAK
DOS image write	A2FC.DOS	A2FC.BAK
Preferences in A2FILE/	A2FILE.TMP	A2FILE.BAK
GOTO in A2FILE/	GOTO.TMP	GOTO.BAK

Target plus backup can mean installation succeeded but cleanup failed. **Backup without target** can mean restoration is needed. A **temporary** can be incomplete or verified but not installed. Preserve the candidates until checked; absence of a backup does not prove an interrupted write completed. A reported successful save with a retained backup is distinct from a failed save. Read the exact message and validate the saved target before cleanup.

For a partial move, compare both sides file by file. **A2MOVE.LST is a work list, not file contents**: do not replay, edit or delete it as a recovery shortcut. Copy missing files into a fresh directory and retain remaining sources. Extraction may leave partial output under its normal filename.

Keep power on for files in **/RAM**; do not reboot or accept AUX reclamation. A2FC does not infer original names, automatically resume interrupted work or promise survival of a power cut during physical metadata writes. More failure cases are documented in Data safety.

Read documents and pictures

Text and documents

T reads text; **H** shows offsets, bytes and characters. Space/Return/Down advances; B/Up goes back; Escape returns. TEXT clips lines at 80 columns and remembers 96 page starts; **R** restarts it. Use **MDVIEW** for wrapped text and Markdown, with 64 previous pages and no forward limit. It also reads extracted Magic Window .MW documents. Teach extended files are unsupported. HEX uses **G** for a seven-digit offset and **R/E** for first/last page.

T lists Applesoft, Integer BASIC and Business BASIC (.BA3) without running them. T/Return reads AppleWorks word-processing text, omitting layout commands. Return on an AppleWorks database or spreadsheet opens **AWDATA**: Space/Down and B/Up page, Left/Right moves columns, R restarts, and F switches between a sheet and cell details. Database TAB shows categories 23–30. Limits are 2,688 records or 882 spreadsheet rows; displayed formulas/results are the saved ones, not a recalculation. Formats, reports and window settings are omitted.

Pictures: Return or I

Return and I select the same recognized picture formats on both CPU editions. Explicit packed types take priority; signatures identify DGR and HGRR/DHRR. An identification read/close error stops opening. **H** explicitly selects hex. Specialized viewers and media tools require **800K or XL**.

Picture or font	Identification / tool
Raw HGR / DHGR, HGRR / DHRR	IMAGE; raw pages or version 1 RLE. Dazzle Draw pictures are raw DHGR.
Lo-res / DGR	DGRVIEW; DGR signature or BIN/FOT with aux \$0400. I can ask for a raw sprite width.
Extasie / Arlequin	EXTASIE (\$F2); ARLEQUIN (\$F8 with its signature).
MacPaint	MACPAINT; .MAC, optionally with a MacBinary header.
Packed FOT / LZ4FH	PACKFOT (\$08/\$4000 or \$4001); LZ4FH (\$08/\$8066).
Packed 816/Paint	PAINT816; BIN with aux \$E001 or \$E002.
Purplesoft	PURPLE; matching .F0T01 and .F0T02 in one directory.
Print Shop	PRINTSHOP; 572/576-byte BIN clip art, aux \$4800/\$5800/\$6800/\$7800.
MGTK or hi-res fonts	FONTVIEW; type \$07, or select the tool for an untyped font.
Applesoft shapes	SHAPES; .SHAPE or choose the tool directly.

Raw HGR accepts **8,184/8,192 bytes**; DHGR accepts **16,376/16,384**, auxiliary plane first: the one-file layout Dazzle Draw saves (BIN, aux \$2000, 16,384 bytes), so its pictures and slide-show disks open as they are. Left/Right browses the previous/next file handled by the same viewer, including across directory windows. At an end it does nothing. Loading shows the filename; Escape returns to the panels. Ordinary small BIN files open in hex with Return; I can treat a suitable one as a sprite.

Viewers that need AUX ask **before** writing it and can rebuild /RAM empty. Consent lasts only for that browsing session; leaving and reopening asks again. Malformed pictures are refused. LZ4FH, PRINTSHOP, FONTVIEW and SHAPES preserve /RAM. Keep both Purplesoft companion files; extended EVE modes require compatible hardware. MACPAINT shows 560 × 192 of its 576 × 720 image; Up/Down pans by 96 lines. SHAPES shows 24 shapes a page, Space/Down next and B/Up previous.

Play music

Music players are included on **800K and XL**, run in the foreground and preserve **/RAM**. **P** pauses/resumes, **Escape** stops, and Left/Right selects the previous/next tune of the same type in the directory. A new track starts unpaused. A missing card or invalid data produces an error instead of playback.

MUSIC: MB1 .MB, up to 4,096 bytes, six Mockingboard voices. **PT3**: up to 65,535 bytes, three voices or standard two-module 02TS TurboSound with six. **DUET**: up to 7,168 bytes, type \$D5/\$D0E7, .ED, or compatible BIN names beginning M. .

Mockingboard is detected automatically. A Mockingboard 4c on //c can hide its mouse firmware: A2FC then keeps keyboard navigation and disables mouse calls until that firmware is restored by reset. A plain //c without the card is unaffected. PT3 keeps its source disk open: leave it mounted. Slow reads and large dual modules can delay playback; 50 Hz is not guaranteed for every tune. Unsupported effects, invalid pointers and I/O errors stop the player.

DUET uses the speaker when no Mockingboard is present. **1/2** switches speaker/card; **D** cycles speaker pulse width. Keys take effect at the next record on the speaker. Files are validated before playback; FIXTYPES can review old extracted DUET files without changing their content.

Extract archives and run programs

Extract into the other panel

Open the destination first, select the archive and choose its tool in !. Return also recognizes common archive suffixes. Existing names are not replaced.

UNSHRINK: .SHK, stored/LZW1/LZW2 data and disk images; unsupported members and resource forks are skipped and reported. **BINARY2**: .BNY/.BQY, whose compressed members may need another extraction. **SCIIBIN**: .BSC/.BSQ with ordered parts and CRC checks. **UNSQ**: .QQ/.ACU, directories skipped and paths flattened. **UNWRAP**: AppleSingle/MacBinary data forks, with omitted resource forks reported.

ShrinkIt requires permission to clear **/RAM** and refuses it as the destination. Escape cancels between blocks. Outputs are read back; errors stop the operation. Cleanup removes only the current new output, where possible; earlier completed files remain. A retained partial file is reported and blocks retry under that name. Preserve it until inspected. Do not assume unsupported or skipped members were recovered; ACU lengths are checked because its stored CRCs are unreliable.

Run a program

Return runs BAS, INT or SYS after confirmation; **X** also runs BIN. The program replaces A2FC and does not automatically return. T lists BASIC without running. 800K/XL include BASIC.SYSTEM and INTBASIC.SYSTEM; keep the selected program online while its runtime loads. BIN must load from \$0800 and end below \$BB00. A failed preference save asks whether to run anyway. The confirmation shows A2FC's full return command: from BASIC's] prompt, reinsert its disk and use that command, or BYE to the ProDOS selector. Shorten BAS paths over 46 characters if launch fails. Integer BASIC's runtime supports only a subset of DOS commands.

Work with disks and images

Browse and extract

Return opens supported ProDOS or DOS 3.3 images as folders: PO, DSK/DO, 2MG and supported DiskCopy 4.2 containers. HFS is not supported. Escape at the root leaves the image. **C** extracts the selection or tags to the other panel. ProDOS extraction supports seedling/sapling files up to 128 KB; enter each subdirectory separately. Existing targets are refused; incomplete outputs are removed where possible and failed cleanup is reported.

A physical DOS 3.3 disk appears in / with slot/drive. **C** extracts to ProDOS on 800K/XL. BIN/BAS/INT headers and sector padding are removed according to the header's exact length; BIN retains its load address, BAS uses \$0801. TXT keeps sector data. Leave through / before opening a changed disk again.

Add or replace files

IMGPUT: open a ProDOS image opposite a selected source and press C, or choose ! → **Disks** → **IMGPUT**. It copies only the cursor file, ignoring tags; V refuses. The image needs a free directory slot and enough free blocks. Files above 128 KB and directory growth are unsupported. Data is verified before allocation metadata is installed; interrupted writes can reserve space.

DOSWRITE: with a real DOS disk opposite, C confirms source name, slot and drive. TXT/BIN/BAS/INT up to 65,535 bytes are supported, one cursor file; existing names are refused. The source must be on another device. **DOSREPL** replaces an existing DOS file using free sectors, verifies it, then switches its catalog entry; allow room for both copies. **DOS33W** deletes or renames after confirming the named file and drive. Locked files and inconsistent allocation are refused.

These tools require 800K/XL and standard 35-track, 16-sector DOS disks on a Disk II in slots 1–7. They check physical protection and allocation before writing. A failed write can leave reserved sectors; preserve the disk before further work. Physical VTOC/catalog writes are not atomic.

For a **DOS image**, C supports DSK/DO and DOS-order 2MG. The tool copies the whole image to A2FC.DOS, validates and modifies that copy, then installs it with A2FC.BAK protecting the old image. Allow space for a second full image. Protected containers and existing recovery names are refused. Keep reported recovery files; the source file is never deleted.

Whole disks, formatting and physical copies

W writes an image to a formatted disk, reads a disk into a new image, or copies disks. PO, DSK/DO and 2MG are supported. Choose the same source/target drive for exchanges. Each prompt identifies SOURCE or TARGET and slot/drive. Writes require **ERASE**, refuse the program disk and verify each block. **F** formats the named drive after ERASE; Escape cancels beforehand. Disk II formatting clears /RAM with consent and refuses to run from /RAM.

NIBCOPY copies standard Disk II tracks with one or two drives at normal 1 MHz speed. Write-protect the source first. Accept AUX use only after saving /RAM, then confirm destruction of **all target files, including locked files**. Single-drive mode requires target confirmation after every exchange. Each source track is read twice and the written track verified; Escape stops between tracks. An incomplete target is not a valid backup. Nonstandard/protected, 13-sector, half-track and weak-bit formats are unsupported; this is not a flux copier.

PASCAL and **CPM** extract supported images into the other panel without modifying the source. Names are adapted to ProDOS; existing targets are skipped. Pascal text retains its header/padding. CP/M data is rounded to 128-byte records; text ends at \$1A. Unsupported directory layouts are refused. Use **IMGCONV** for container conversion, not extension changes. No physical-write operation promises rollback through a power failure.

Find and maintain files

Press **!**, choose a category, then a tool. Return opens; Escape goes back. Up/Down moves one entry, Left/Right six, and an initial jumps by name. Categories describe tasks; no category disk swap is needed. Except **COMPARE**, **VERIFY** and **MOVE**, the tools below require **800K or XL**. Read each result, especially partial scans, skipped files and failed cleanup.

Files and navigation

- **COMPARE / CRC / VERIFY**: compare opposite namesakes byte for byte; calculate CRC-32; read files or volume blocks and report errors.
- **TAGPAT / SEARCH / FIND**: tag by pattern; search current-directory text; search a volume by name or contents (" prefix). **FIND**: TAB filters, N pages, Return jumps and V shows text matches.
- **GOTO**: P opens a path; A adds a favourite, D + digit removes, M + two digits reorders, 1–9 jumps.
- **RENAME / DATE / TXTCONV**: batch names; modification dates; CR/LF/CRLF, high-bit, tabs or UTF-8 accent conversion. Conflicting names are preserved.
- **IDENT / FIXTYPES**: identify formats; confirm attribute changes individually. Hints do not prove a damaged file is valid.
- **SYNC / TREE**: copy missing/newer files recursively; show directory sizes.

TAGPAT uses = for any string and ? for one character; comma filters include T04, >2000, <2000 and D (today). T adds tags, U removes, X replaces. **GOTO** paths are at most 63 characters. **DATE** accepts DDDMMYYYYHHMM, 1940–2039; creation dates remain unchanged. **SYNC** preserves destination-only and newer files, refuses overlapping trees and stops if replacement/recovery fails. An unknown source date does not authorize replacement. Keep A2FC. SYNC/A2FC. BAK.

Volumes and recovery

VOLINFO audits space, allocation and fragmentation: M bitmap, F file blocks, E export, N/P pages. Partial counts are unconfirmed; only a complete export ends with END REPORT. **VOLNAME** renames a volume and updates affected paths; a name already online is refused. **MKIMAGE** creates an empty PO/2MG data volume, without a boot program, up to 32,767 blocks.

FIXIT only reads. It reports ProDOS directory/allocation faults and refuses incomplete scans rather than declaring lost blocks. For volumes above 4,096 blocks, Q checks directories only; F performs a full scan requiring AUX consent. A quick clean verdict is not a full allocation check. It does not check DOS disks.

REPAIR shows a plan and requires **F**, then **FIX**, before writing. It refuses the program volume, cross-links and uncertain scans; some damaged files also prevent freeing unclaimed blocks. Writes are read back, restoration is attempted on error, and the volume is rescanned. Backups are only in RAM: interruption can leave a partially applied plan. Large scans can take minutes.

UNDELETE recovers validated deleted files to another volume without changing the source. Reused/ambiguous allocation, deleted directories and resource forks are refused. **RESCUE** retries reads and zero-fills unreadable chunks, recording losses in BASE.LOG. File output is BASE.REC; volume parts BASE.P01, .P02, etc. must be concatenated in order. Partial output remains. Inspect the log and recovered contents: neither tool guarantees recovery of missing data.

WIPE F checks allocation before zeroing free blocks; **W** destroys an entire volume after ERASE, excluding the running program's volume. **BOOTBLK** copies boot blocks after confirmation, with readback and RAM-only originals for attempted restoration. **BLKVIEW** reads blocks and can extract a range; **DISASM** reads 6502/65C02 instructions and exports text. Their exports can remain partial after an error. None of these tools should be tried first on an irreplaceable disk.

Limits, troubleshooting and VDrive

Read the status before acting

Long copies distinguish **Copying** from **Verifying**; the verification counter starts again at zero. Every long tool shows a bar or, where none fits, turns a / \ mark at the end of the line above the messages (DOS3.3: the last cell of the key bar, which also turns inside DOS's own long calls: the format, and the retries on a diskette that was never formatted). A drive change or a head recalibration can still hold it for a second or so. Do not remove media just because a counter has stopped; record the screen and wait for a result or a safe cancellation point.

Escape cancels where the active tool permits it. Completed work remains and partial output may require recovery. A result that reports skipped, unsupported, unreadable or retained files is not a fully successful batch. Check the actual message before retrying or cleaning up.

Symptom	Action
Startup refusal	Choose the matching CPU/ROM edition; ProDOS needs 128 KB and 80 columns.
Missing/stale tool	Use 800K/XL and matching program/plugins.
Old catalog or a large directory	Ctrl-R rereads; cross window edges beyond 139 entries. Large windows use disk order.
Deep traversal refused	Work on smaller subdirectories; an incomplete tree must not be deleted.
Temporary/backup blocks retry	Preserve candidates and follow incident recovery.
Run or image opening fails	Check paths, runtime, type and actual container; renaming extensions does not convert.

Boundaries to keep in mind

A matching size/date does not prove a recovered file complete: COMPARE checks bytes, M only compares size/date. Unsupported forks or storage may be refused or skipped with a report. AUX consent clears /RAM without backing it up. Verification cannot make physical metadata writes atomic across a power cut.

VDrive: remote volumes

A detected Super Serial Card or //c serial port can expose two remote ProDOS volumes at **115,200 bps**. VDrive tries slot 2 (the //c's modem port) first, then slots 3 to 7, and **never slot 1**: that is the printer's (a //e's printer SSC, the //c's printer port), and VDrive would reprogram it and send it its packets. It does not even read that card. A VDrive cable on a slot-1 card is therefore not served: put the card in slot 2. In slots 2 to 7, a Super Serial Card whose mode switches (SW1-5 and SW1-6) are not set to communications mode -- printer mode, or a SIC emulation -- is a printer's: VDrive reads those switches first and skips the card without writing to it. Set the card that talks to the VDrive host to communications mode. A //c has no switches, so nothing tells a printer on its modem port from a modem: port 2 is always taken when present, and a printer plugged there receives VDrive's packets. Keep printers off the //c's modem port while A2FC runs. Use ADTPro's virtual-drive server, `veserver.py` or `sur1-server`. The status line identifies the serial interface and assigned drives; browse and copy as with local volumes. The host supplies date/time during reads. A disconnected server produces an I/O error; it is not an empty or absent file. The driver is removed on quit or program launch.

For reproducible failures, note the A2FC version/edition, machine, disk type, full paths, exact message and actions. Keep a disk image before any repair. The project's issue tracker and detailed data-safety notes are linked in the last chapter. Automated emulator tests do not replace trials on physical drives.

Credits and further reading

A2 File Cmd is **GNU GPL v3** software by **Arnaud Verhille**. Its two-panel interface draws inspiration from A2Command and Norton Commander. Code provenance, source notices and redistribution terms remain in the repository and LICENSE.

Acknowledgements

- **cc65 / Oliver Schmidt**: toolchain and Apple II startup sources.
- **Colin Leroy, a2tools / Ammonoid**: serial virtual-drive approach and glue.
- **Jerry Hewett and Gary Desrochers**: public-domain Hyper-FORMAT routines, carried by ADTPro and adapted for Disk II formatting.
- **Apple ProDOS references**: filesystem, MLI, paths and allocation rules.
- **Paul Lutus**: Electric Duet format and original player; **Alex Patalenski** and **Emil Dotchevski**: improved speaker player and its published listing.
- **Cybernesto, electric-mock**: DUET's Mockingboard rendition; **Vince Weaver, pt3_lib**: PT3 player work adapted by A2FC.
- **Andy McFadden, CiderPress II**: format notes and real reader test samples.
- **NuLib / ShrinkIt references**: NuFX and Binary II specifications.
- **David Schmidt, ADTPro**: disk-transfer and virtual-drive workflows.
- **POM2 / Arnaud Verhille**: separate emulator for repeatable tests.

The complete attributions, reference URLs and source-file notices are preserved in Credits and provenance. Keep that document, the affected source notices and LICENSE when redistributing a modified build. DEMO/README on the XL disk tells generated demonstrations from the CiderPress II samples it reuses; data/CP2/README.TXT describes that corpus.

Where to go next

- **After an incident**: open RECOVER on the program disk and follow this guide's recovery chapter before retrying or deleting anything.
- **Failure coverage**: Data safety describes temporary names, retained originals, error handling and physical-write limits.
- **Real machines**: Hardware checklist covers disposable test media and records the actual machine/drive results.
- **DOS3.3 internals**: Developer guide holds implementation, build and timing details removed from this concise user guide.
- **Media**: Sample media documents test material and sources.
- **Builds and extensions**: the repository README, sdk/README.md and bench/README.md describe development and test workflows.
- **Bugs and source**: A2 File Cmd on GitHub.

This is the guide for **0.9.4**. It describes supported behavior and limits; it is not a claim that physical hardware qualification has finished. Keep original disks safe, use matching program/tools, and read each operation's result before starting the next one.